Overview of K-means clustering

Subject: K-means clustering

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

1.

The most common algorithm uses an iterative refinement technique. Due to its ubiquity, it is often called "the k-means algorithm"; it is also referred to as Lloyd's algorithm, particularly in the computer science community. It is sometimes also referred to as "naïve k-means", because there exist much faster alternatives. Given an initial set of k means m1(1), ..., mk(1) (see below), the algorithm proceeds by alternating between two steps:

2.

| S i | {\displaystyle |S_{i}|} is the size of S i {\displaystyle S_{i}} , and ■ · ■ {\displaystyle \|\cdot \|} is the usual L2 norm . This is equivalent to minimizing the pairwise squared deviations of points in the same cluster:

3.

Modern astronomical surveys like APOGEE and the Gaia catalog collect spectroscopic and photometric data for millions of stars, making manual classification impractical at scale. K-means clustering has been applied to this data to automatically identify distinct stellar populations based on properties like luminosity, temperature, color, and chemical abundance. One major application is identifying stellar populations through chemical tagging. Stars that form together in the same cluster share similar chemical compositions, so clustering algorithms applied to abundance data can recover these groupings even after the stars have spread throughout the galaxy. Studies using k-means on APOGEE infrared spectra measured abundances of up to 13 elements and successfully distinguished known star clusters from surrounding field stars. This has also revealed populations with unusual abundance patterns that would be difficult to catch through manual inspection. K-means has also been combined with dimensionality reduction techniques to classify large numbers of unlabeled survey objects by separating stars, galaxies, and quasars based on their observed light properties. Labels from confirmed examples are then propagated to nearby cluster members. This approach has achieved accuracy comparable to fully supervised methods while requiring far fewer labeled examples, which makes it well suited for the scale of modern sky surveys.

4.

Recent developments Recent advancements in the application of k-means clustering have explored the integration of k-means clustering with deep learning methods, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to enhance the performance of various tasks in computer vision, natural language processing, and other domains.

5.

Assignment step Hartigan and Wong's method starts by partitioning the points into random clusters { S j } j ∈ { 1 , ■ k } {\displaystyle \{S_{j}\}_{j\in \{1,\cdots k\}}} . Update step Next it determines the n , m ∈ { 1 , … , k } {\displaystyle n,m\in \{1,\ldots ,k\}} and x ∈ S n {\displaystyle x\in S_{n}} for which the following function reaches a maximum Δ (m , n , x) = ϕ (S n) + ϕ (S m) − ϕ (S n ■ { x }) − ϕ (S m ∪ { x }) . {\displaystyle \Delta (m,n,x)=\varphi (S_{n})+\varphi (S_{m})-\varphi (S_{n}\setminus \{x\})-\varphi (S_{m}\cup \{x\}).} For the x , n , m {\displaystyle x,n,m} that reach this maximum, x {\displaystyle x} moves from the cluster S n {\displaystyle S_{n}} to the cluster S m {\displaystyle S_{m}} . Termination The algorithm terminates once Δ (m , n , x) {\displaystyle \Delta (m,n,x)} is less than zero for all x , n , m {\displaystyle x,n,m} . Different move acceptance strategies can be used. In a first-improvement strategy, any improving relocation can be applied, whereas in a best-improvement strategy, all possible relocations are iteratively tested and only the best is applied at each iteration. The former approach favors speed, whether the latter approach generally favors solution quality at the expense of additional computational time. The function Δ {\displaystyle \Delta } used to calculate the result of a relocation can also be efficiently evaluated by using equality

6.

The relaxed solution of k-means clustering, specified by the cluster indicators, is given by principal component analysis (PCA). The intuition is that k-means describe spherically shaped (ball-like) clusters. If the data has 2 clusters, the line connecting the two centroids is the best 1-dimensional projection direction, which is also the first PCA direction. Cutting the line at the center of mass separates the clusters (this is the continuous relaxation of the discrete cluster indicator). If the data have three clusters, the 2-dimensional plane spanned by three cluster centroids is the best 2-D projection. This plane is also defined by the first two PCA dimensions. Well-separated clusters are effectively modelled by ball-shaped clusters and thus discovered by k-means. Non-ball-shaped clusters are hard to separate when they are close. For example, two half-moon shaped clusters intertwined in space do not separate well when projected onto PCA subspace. k-means should not be expected to do well on this data. It is straightforward to produce counterexamples to the statement that the cluster centroid subspace is spanned by the principal directions.

7.

The primary drawback of standard k-means++ is its inherent sequential nature. To find k initial centers, the algorithm must make k separate passes over the data, as the selection of each new center depends on the previous choices. This becomes prohibitively slow when clustering massive datasets into a large number of clusters. To overcome these limitations, researchers developed k-means||, a parallel version of k-means++. Instead of sampling a single point per pass, k-means|| uses an oversampling factor ■ = Ω (k) to sample multiple points in each round. The algorithm drastically reduces the number of passes required, obtaining a nearly optimal set of centers with a time complexity of O(log n). In practice, as few as five rounds are needed to reach a high-quality solution. After several rounds, the algorithm typically has O(log n) intermediate centers. These are then assigned weights based on how many points are close to them and re-clustered, often using standard k-means++, to reduce the final set to exact k centers.

8.

a r g m i n S ■ ∑ i = 1 k ∑ x ∈ S i ■ x − μ i ■ 2 = a r g m i n S ■ ∑ i = 1 k | S i | Var ■ S i

$${\displaystyle \mathop {\operatorname {arg\,min} } _{\mathbf {S} }\sum _{i=1}^{k}\sum _{\mathbf {x} \in S_{i}}\left\|\mathbf {x} -{\boldsymbol {\mu }}_{i}\right\|^{2}=\mathop {\operatorname {arg\,min} } _{\mathbf {S} }\sum _{i=1}^{k}|S_{i}|\operatorname {Var} S_{i}}$$